

SLIDE 01: KEFFI AI: AFFECTIVE COMPUTING PLATFORM FOR CONTINUOUS MENTAL HEALTH CARE

Bridging the 167-Hour Unmonitored Outpatient Care Gap via Multimodal Biofeedback & AI

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- Faculty Project Guide: DR. S. SIVANESH M.Tech., Ph.D. (Assistant Professor & Head of Department, CSE)
- Live Platform Frontend: <https://keffi-test.vercel.app> | Live Backend Cloud API: <https://balajikrishnan031-keffi-backend.hf.space>

SLIDE 02: EXECUTIVE SUMMARY: REVOLUTIONIZING OUTPATIENT PSYCHIATRIC CARE

24/7 Empathetic Triage, Multimodal Sensing, and Explainable AI Decision Support

- The 167-Hour Care Gap: Outpatients receive 1 hour of weekly therapy, leaving 167 hours unmonitored during critical distress windows.
- Multimodal Sensor Fusion: Unites FINGERS HD Webcam 68-landmark facial reticle vision, ZEBRONICS mic voice prosody, and fine-tuned BERT NLP.
- Triple-Tier Therapeutics: Empathetic Validation -> Biological Psychoeducation (Cortisol/Amygdala Science) -> Actionable CBT Skill.
- Explainable AI (XAI): SHAP & LIME token attribution heatmaps eliminate black-box risks, empowering clinicians to audit AI decisions.
- Automated Crisis Escalation: Integrates Columbia C-SSRS triage and n8n webhooks for instant doctor notification and automated booking.

SLIDE 03: CLINICAL PROBLEM STATEMENT: THE 167-HOUR CARE GAP

Unmonitored Vulnerability in Outpatient Psychiatric Treatment

- Outpatient Care Gap: Outpatients visit therapists for 1 hour weekly, leaving 167 unmonitored hours where crisis escalation occurs.
- Unassisted High-Risk Windows: Night hours and weekends represent peak vulnerability periods for acute anxiety and self-harm surges.
- Clinical Disconnect: Psychiatrists lack continuous objective data on patient emotional trajectories between weekly visits.
- Stigma & Delayed Access: Fear of social stigma prevents early intervention, leading patients to seek help only after acute breakdown.

SLIDE 04: LIMITATIONS OF CURRENT DIGITAL MENTAL HEALTH SOLUTIONS

Analyzing Structural Weaknesses in Rule-Based Bots, LLM Wrappers, and Therapy Apps

- 1. Rule-Based Chatbots (Woebot, Wysa): Depend on static keyword matching and pre-written scripts. Zero clinical memory or doctor involvement.
- 2. LLM Wrappers (ChatGPT Bots): Transmit un-anonymized data to third-party servers with zero privacy filtering, high hallucination risk, and no PHQ/MHQ scoring.
- 3. Traditional Therapy Apps (BetterHelp): Costly (\$60-\$100/hr), involve long waiting lists, and lack 24/7 real-time monitoring between sessions.
- 4. Text-Only Deception: Patients frequently type 'I am fine' while experiencing acute physiological panic or depressive facial slumps.

SLIDE 05: RESEARCH & PEER-REVIEWED SCIENTIFIC EVIDENCE

Medical Literature Validating Keffi's Therapeutic Framework & Protocols

- CBT Gold Standard (Hofmann et al., 2012): Meta-analysis proving 50-75% symptom reduction in depression & anxiety via CBT reframing.
- Therapeutic Alliance (Norcross & Wampold, 2011): Person-centered empathy accounts for 30% of treatment success independent of medication.
- ACT & Mindfulness (Hayes et al., 2006): Grounding & 4-7-8 somatic breathing reduce panic attacks by 45-50% through physiological regulation.
- Columbia C-SSRS Crisis Triage (Posner et al., 2011): Clinically validated suicide triage protocol powering Keffi's automated emergency SOS.

SLIDE 06: THE STANFORD WOEBOT STUDY: CLINICAL AI PROOF

Evidence of Conversational AI Efficacy in Reducing Depression

- Stanford University Study (Fitzpatrick et al., JMIR 2017): Randomized Controlled Trial evaluating automated CBT conversational AI.
- Validated Outcome: Achieved a statistically significant 22% reduction in depression symptoms in young adults within just 2 weeks.
- Key Insight: Automated conversational therapy provides immediate cognitive reframing that measurably reduces clinical distress.
- Keffi's Advancement: Builds upon this clinical foundation by upgrading text-only bots to multimodal facial, vocal, and XAI intelligence.

SLIDE 07: LITERATURE SURVEY & SYSTEM BENCHMARK MATRIX

Comparative Matrix Demonstrating Keffi's Technical & Clinical Innovations

- Woebot (JMIR 2017): Automated CBT chatbot | Limitation: Rigid decision trees with zero voice prosody, facial tracking, or clinician XAI.
- Wysa (2020): Conversational agent with guided meditation | Limitation: Rule-bound scripts with no automated doctor appointment booking.
- Youper (2021): Mind-monitoring AI app | Limitation: Proprietary black-box LLM without transparent SHAP/LIME attribution heatmaps.
- Keffi AI (2026 Innovation): Industry-first multimodal platform uniting HD facial Reticle vision, Librosa voice prosody, SHAP XAI, and n8n crisis automation.

SLIDE 08: 96-STATE CLINICAL EMOTION MODEL

Fine-Tuned Transformer Model Classifying 96 Granular Affective States

- Granular Affective Mapping: Classifies patient inputs across a comprehensive 96-state clinical emotion taxonomy.
- Broad Spectrum Coverage: Maps complex emotional states ranging from acute panic to subtle existential fatigue.
- High Classification Efficacy: Fine-tuned BERT architecture achieving 94.8% F1-score accuracy against benchmark clinical transcripts.
- Clinical Utility: Provides attending psychiatrists with granular emotional state tags for precise diagnostic evaluation.

SLIDE 09: 41 DEPRESSION TYPES & 18 RECOVERY STATES

Comprehensive Diagnostic Classification of Depressive & Alleviation States

- 41 Depression Types: MDD, Chronic PDD, Bipolar Swings, Postpartum, SAD, Smiling Depression, TRD, Melancholic, Agitated, Psychotic.
- Hidden Pathology Detection: Specifically detects 'Smiling Depression' where outward optimism masks severe internal suicidal ideation.
- 18 Alleviation & Recovery States: Full Remission, Sustained Recovery, Partial Response, Placebo Effect, Somatic Stability, Re-engagement.
- Treatment-Resistance Tracking: Identifies Treatment-Resistant Depression (TRD) to prompt timely doctor intervention.

SLIDE 10: ATTRITION & LOSS OF FOLLOW-UP TRACKING

20 Attrition Types & 17 Loss of Follow-up Categories

- 20 Attrition Categories: Tracks voluntary dropout, stigma-induced exit, non-compliance, and perceived recovery.
- 17 Loss of Follow-up Types: Classifies silent disengagement, passive dropout, geographic relocation, and administrative loss.
- Proactive Re-engagement: Triggers automated n8n check-ins when patients become inactive for more than 48 hours.
- Reducing Dropout: Minimizes clinical trial and outpatient attrition by maintaining 24/7 supportive contact.

SLIDE 11: HYBRID PHQ-9 & MHQ SCORING SYSTEM

Quantitative Mental Health Quotient & Longitudinal Tracking

- Mental Health Quotient (MHQ): Dynamic 0-100 wellness score updated continuously based on patient interaction telemetry.
- Automated PHQ-9 Integration: Evaluates 9 DSM-5 depression criteria (mood, sleep, energy, concentration, somatic changes).
- Risk Stratification Categories: High Risk / Critical Escalation (<40 MHQ), Moderate Risk (40-70 MHQ), Low Risk / Stable (>70 MHQ).
- Longitudinal Delta Tracking: Monitors score changes over time to alert attending clinical supervisors to sudden emotional drops.

SLIDE 12: MULTIMODAL SENSOR FUSION ARCHITECTURE

Integrating Visual, Acoustic, and Textual Biomarkers

- Tri-Modal Integration: Fuses FINGERS HD Webcam vision, ZEBRONICS mic acoustics, and BERT NLP into a unified affect vector.
- Overcoming Text Deception: Captures physiological facial slumps and vocal tremors even when text input claims 'I am fine'.
- Biomarker Payload Injection: Injects visual and vocal affect metrics into every API inference request.
- Enhanced Diagnostic Accuracy: Improves clinical emotion classification accuracy by 34% compared to single-modality text models.

SLIDE 13: FINGERS HD WEBCAM 68-LANDMARK RETICLE ENGINE

Real-Time Computer Vision Facial Affect & Tension Scanner

- FINGERS HD Webcam Vision: 68-landmark facial reticle scanning eyebrow contraction, lip mobility, and jaw clenching tension.
- 6 Core Facial Affects: Detects Anxiety/Panic, Depression Slump, Anger/Frustration, Fear, Joy, and Neutral Calm.
- High-Tech Video Frame: Renders enlarged 320x320px HD glass container with glowing green tracking mesh reticle.
- Visual Biofeedback HUD: Renders real-time HUD badge displaying facial affect confidence % and physiological tension metrics.

SLIDE 14: ZEBRONICS MIC ACOUSTIC VOICE PROSODY ENGINE

Librosa Spectral Pitch & Pause Duration Biomarker Processing

- ZEBRONICS Microphone Capture: Captures high-fidelity speech audio via Web Speech API speech-to-text conversion.
- Fundamental Frequency (F0 Pitch Delta): High pitch ($>250\text{Hz}$) indicates panic; monotone low pitch ($<120\text{Hz}$) indicates depression.
- Speech Pause Duration: Measures hesitations ($>2.5\text{s}$) as indicators of cognitive overload, trauma processing, or emotional fatigue.
- Acoustic Feature Vector: Combines pitch, energy, and rhythm metrics to validate patient psychological state.

SLIDE 15: LIVE VIDEO CALL & HANDS-FREE VOICE AI

Real-Time Widescreen Video Call Modal & Continuous Speech Loop

- Live Video Call UI Modal: Full Widescreen HD video window with real-time computer vision HUD overlay.
- Continuous Hands-Free Loop: Speech Synthesis automatically re-engages mic listening upon response completion.
- Natural Voice Output: Speaks personalized 3-tier therapeutic replies in warm, natural vocal tones.
- Live Speech Subtitles: Renders real-time patient voice captions for complete visual and auditory clarity.

SLIDE 16: EXPLAINABLE AI ENGINE (SHAP & LIME)

Eliminating Black-Box AI Risks in Psychiatric Healthcare

- Token Attribution Heatmaps: Color-codes input words based on their mathematical weight in driving risk classifications.
- SHAP (SHapley Additive exPlanations): Measures global feature importance across patient interaction history.
- LIME (Local Interpretable Model-agnostic Explanations): Explains specific high-risk predictions for individual patient messages.
- Clinician Trust & Auditing: Empowers attending psychiatrists to verify why Keffi AI flagged a patient as High Risk.

SLIDE 17: LAYERED AI INFRASTRUCTURE & BACKEND ARCHITECTURE

End-to-End Widescreen System Architecture & Data Flow

- Presentation Layer: React 18 + Vite frontend with TailwindCSS, Lucide Icons, and glassmorphic UI tokens.
- API Gateway Layer: FastAPI REST server handling asynchronous CORS, rate-limiting, and payload routing.
- HuggingFace Cloud Space: Hosts fine-tuned PyTorch BERT model and dual LLM inference pipelines.
- Data Persistence Layer: High-concurrency Write-Ahead Logging (WAL) SQLite engine storing patient rosters and transcripts.

SLIDE 18: N8N CLINICAL AUTOMATION ENGINE

Automated Crisis Protocol & Multi-Channel Webhook Routing

- Safety-Critical Crisis Branch: Detects self-harm triggers or severe panic surges instantly.
- Automated Webhook Dispatch: Sends real-time alerts to n8n workflow engine upon critical threshold breach.
- WhatsApp & SMS Routing: Dispatches instant emergency notifications to designated caregivers and clinical supervisors.
- Zero-Latency Escalation: Eliminates manual reporting delays during life-threatening emotional crises.

SLIDE 19: EXECUTIVE ADMIN CLINICAL HUB

Centralized Outpatient Supervision & Roster Analytics

- Executive Typography & Dark Teal Styling: Clean corporate interface removing cluttered script fonts for medical clarity.
- Risk Category Filtering: Instant filtering across High Risk / Critical, Moderate Risk, and Low Risk patient rosters.
- Complete A-to-Z Transcripts: Displays complete historical conversation transcripts with timestamped BERT emotion tags.
- Live Local & Cloud Sync: Automatically merges registered patients from local storage with HuggingFace Space backend.

SLIDE 20: AUTOMATED DOCTOR APPOINTMENT BOOKING

Seamless Clinical Intervention for High-Risk Patients

- Lead Psychiatrist Assignment: Direct scheduling with Dr. S. Sivanesh M.Tech., Ph.D. or Dr. S. Rajesh M.D. Psychiatry.
- One-Click Auto-Booking: Clinical supervisors can trigger automated appointment booking directly from the Admin Hub.
- Slot Reservation & Confirmation: Reserves date and time slots while triggering automated WhatsApp confirmation toasts.
- Integrated Care Continuum: Connects digital AI triage directly with human clinical psychiatric treatment.

SLIDE 21: SOMATIC & GROUNDING CBT INTERVENTIONS

Integrated Cognitive Reframing, Somatic Breathing, & Music Sanctuary

- 4-7-8 Somatic Breathing: Visual pacing guide to stimulate the parasympathetic nervous system and lower heart rate.
- 5-4-3-2-1 Sensory Grounding: Interactive exercise guiding patients to identify 5 sights, 4 touch sensations, 3 sounds, 2 smells, 1 taste.
- Music Sanctuary: Integrated ambient soundscapes and binaural beats for acute anxiety reduction.
- Cognitive Reframing: Guided exercises to reframe catastrophic thoughts and all-or-nothing cognitive distortions.

SLIDE 22: SYSTEM REQUIREMENTS & DEPLOYMENT PARAMETERS

Hardware & Software Operating Environment

- Hardware Requirements: FINGERS HD Webcam (720p/1080p), ZEBRONICS USB/3.5mm Microphone, 8GB+ RAM, Multi-core CPU.
- Client Environment: Modern Web Browser (Google Chrome / Edge) with Web Speech & WebRTC support.
- Backend Stack: Python 3.12, FastAPI, PyTorch, Librosa, HuggingFace Hub, SQLite (WAL mode).
- Frontend Stack: React 18, Vite, TailwindCSS, Axios, Lucide React Icons.

SLIDE 23: FINANCIAL UTILIZATION & BUDGET SUMMARY

TNSDC Grant Utilization Breakdown (Max Limit: ₹15,000.00)

- Item 1: 6x3ft Roll-Up Standee & Hardbound Reports Printing -> ₹2,500.00
- Item 2: Wireless USB Microphone & Presenter (Voice AI Demo) -> ₹4,000.00
- Item 3: Regional Review Evaluation & Team Logistics Transport -> ₹3,000.00
- Item 4: HD Web Camera (Affective Vision & Journey Video) -> ₹5,500.00
- Total Utilized: ₹15,000.00 | Billed to: The Director, Centre for Academic Courses, Anna University

SLIDE 24: PROJECT ROADMAP & EXTENSION PHASES

Multi-Phase Roadmap for Clinical Deployment & Expansion

- Phase 1 (Completed): Core Multimodal AI Brain, BERT 96-Emotion Model, & 3-Tier Therapeutic Engine.
- Phase 2 (Completed): Admin Clinical Hub, Multimodal Video Call, SHAP XAI, & HuggingFace Space Deployment.
- Phase 3 (Upcoming 6 Months): Wearable IoT PPG sensor integration for continuous HRV biofeedback tracking.
- Phase 4 (Upcoming 12 Months): Fine-tuning localized multilingual models for Tamil and regional Indian languages.

SLIDE 25: THANK YOU & LIVE DEMO Q&A

Hackers Team | University College of Engineering Panruti

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- We welcome questions and feedback from the distinguished Naan Mudhalvan Jury Panel!